

Silent Poisoning: A Stealthy and Unified Dual-Space Framework for VFL Backdoor Attacks

Qiao Li , Xiaoya Ma , Zijun Zhang , Jing Chen , Kun He , Ruiying Du , Zijian Zhang , Cong Wu ,
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Abstract—Vertical federated learning (VFL) allows multiple parties to collaboratively train models while keeping their local features private, but its robustness against backdoor attacks remains poorly understood. Existing backdoor techniques developed for horizontal federated learning are largely inapplicable to VFL, as adversaries in VFL cannot access labels and are strictly limited to manipulating local feature representations under server-side supervision. This raises a fundamental question: *can effective and stealthy backdoor attacks be mounted in VFL under such constraints?* In this article, we provide a systematic investigation of backdoor attacks in VFL by exploring attack surfaces in both the input space and the latent space. We first propose an input-space attack that exploits gradient information exchanged during training to subtly perturb local inputs and reshape input–label correlations, thereby bypassing label inaccessibility without relying on surrogate models. We then design a latent-space attack that adversarially steers local embeddings toward target-class representations. To ensure practical undetectability, we incorporate a unified stealthiness regularization strategy into the framework. To quantify this property, we introduce a VFL-specific metric that measures statistical consistency, guiding the trade-off between attack effectiveness and detectability. Extensive experiments show that our attacks achieve high success rates while remaining difficult to detect, exposing previously overlooked security vulnerabilities in VFL systems.

Index Terms—Vertical federated learning, backdoor attack, input-space, latent-space, stealthiness.

I. INTRODUCTION

FEDERATED learning (FL) enables multiple parties to jointly train a model without sharing raw data. Vertical federated learning (VFL) [1] targets a common real-world setting in which participants hold different feature subsets of the same samples. This setting frequently arises in cross-institutional finance and healthcare applications [2]. VFL typically follows a

split-model training protocol. Each party exchanges only intermediate representations and gradients, rather than raw features or complete model parameters. This design improves data privacy but also creates a security landscape that differs fundamentally from horizontal FL.

Research Gap: In VFL, backdoor attacks pose a subtle yet serious threat to model integrity. A malicious participant may inject a hidden trigger into its local feature space during training. When the trigger appears, the jointly trained model produces attacker-chosen outputs, while maintaining normal performance on clean inputs. Unlike attackers in horizontal FL, VFL adversaries face strict constraints. They observe only partial features, have limited or no access to labels, and must follow the training protocol. These restrictions render many existing HFL backdoor strategies [6] ineffective. Despite the increasing adoption of VFL in security-critical domains, the feasibility and risk boundaries of backdoor attacks under such realistic constraints remain unclear. This leads to a central question: *Can VFL systems still suffer effective and stealthy backdoor attacks under strict adversarial limitations?*

Prior VFL backdoor attacks rely heavily on surrogate models for label inference [3], [4], [5]. However, this strategy succeeds only when the server-side top model is structurally simple, such as a fully connected network. In practice, VFL systems often use complex, proprietary architectures. The non-linear dynamics of these models make accurate approximation difficult, which significantly reduces the effectiveness of surrogate-based attacks. Beyond their dependence on surrogate modeling, current VFL backdoor studies exhibit two notable structural shortcomings. One line of work confines adversarial manipulation to a single space, either the input space or the latent space [7]. Such single-space designs are typically analyzed in isolation, without a unified perspective on the two protocol-exposed interfaces inherent in split-model training. As a consequence, the relationship between different attack surfaces remains insufficiently characterized under a common design framework. Another limitation lies in the treatment of stealthiness. In most prior studies, stealth is not explicitly enforced during optimization, but rather arises implicitly as a by-product of attack construction, often resulting in observable distributional shifts in server-side representations [4], [5], [8], [9]. Together, these limitations highlight a clear research gap. We currently lack a unified framework that is surrogate-free, stealth-aware, and effective under strict VFL protocols. Table I summarizes representative attacks and shows that prior methods fail to satisfy these properties,

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TABLE I
COMPARISON OF REPRESENTATIVE VFL BACKDOOR ATTACKS

Method	SF	CSC	SR	Attack Surface
Naseri et al. [5] (2024)	○	○	○	Input
Gu et al. [3] (2023)	○	○	○	Latent
Chen et al. [8] (2023)	●	○	○	Input
Shen et al. [9] (2025)	●	○	○	Input
Ours	●	●	●	Input & Latent

* ● = Supported; ○ = Unsupported.

SF (Surrogate-Free): Establishes backdoor without training a local surrogate model.

CSC (Cross-Space Capability): Instantiates attacks in both input and latent spaces uniformly.

SR (Stealth Regulation): Explicitly constrains malicious representations within the benign manifold.

motivating our proposed approach. Specifically, Table I compares our framework with representative VFL backdoor attacks. Our method is fully surrogate-free and supports cross-space manipulation, while prior works operate in only one space. In addition, we explicitly enforce stealth constraints to preserve statistical consistency with benign distributions, which prior methods do not consider.

Our Proposed Framework: To address these gaps, we propose a unified clean-label backdoor framework for VFL. Our approach exploits inherent weaknesses in split-model training. In a standard VFL setting, the adversary interacts with the system only through protocol-exposed interfaces. Specifically, the adversary can modify local input features and control the latent representations sent to the server. Since the server-side model acts as a closed box and labels remain strictly isolated, the input space and the latent space constitute the complete attack surface for a protocol-compliant participant. Any attempt to manipulate the model outside these interfaces requires protocol violations, such as label inference or unauthorized access. Such violations are impractical in real-world VFL deployments. Therefore, an effective VFL backdoor must satisfy two requirements: (i) *Surrogate-free autonomy*, where the attack operates directly through the exposed interfaces without reliance on surrogate models; and (ii) *Stealth regulation*, which forces adversarial signals to remain statistically consistent with the benign data distribution.

Guided by these principles, we realize our framework through two tightly coupled yet complementary attack mechanisms that operate on distinct stages of the VFL training pipeline. We first consider the input space, where the adversary perturbs its local features to exploit the gradient-coupling effect inherent in split-model training. Although the adversary lacks access to global labels, these optimized perturbations implicitly steer the shared gradients during backpropagation. This allows the model to learn stable backdoor associations while remaining fully protocol-compliant. This mechanism demonstrates that indirect influence via gradients is sufficient for effective backdoor injection. Building upon this insight, we further extend the attack to the latent space by directly manipulating the representations transmitted to the server. Since these representations drive server-side inference, this mechanism eliminates the need for surrogate models. It also works effectively across diverse and complex architectures. Compared to input-space manipulation,

the latent-space attack offers a more direct and architecture-agnostic strategy for establishing backdoor behavior.

Taken together, these two mechanisms form a unified attack framework. The input-space attack enables fine-grained influence under strict protocol constraints. Meanwhile, the latent-space attack exploits the representation interface to ensure robust effectiveness. Both mechanisms operate without surrogate models. Furthermore, we apply an explicit stealth regulation strategy to both approaches. Unlike prior methods that treat stealthiness as a side effect, we formulate it as a primary design objective. We constrain adversarial perturbations to remain statistically consistent with the benign data distribution. This unified regulation establishes a common standard for optimization and evaluation across both attack surfaces. Overall, our framework provides a practical perspective on stealthy, clean-label backdoor attacks in VFL. We demonstrate that an attacker can achieve both high effectiveness and low detectability under realistic protocol constraints.

Our main contributions are summarized as follows:

- We propose a unified, surrogate-free framework for stealthy clean-label backdoor attacks in VFL, systematically identifying the input and latent spaces as complementary and protocol-compliant attack surfaces.
- We design dual-space attack mechanisms that achieve high attack success rates while preserving benign model utility, demonstrating effective manipulation under strict VFL constraints.
- We introduce a VFL-specific stealthiness metric that explicitly enforces statistical consistency between poisoned and benign samples, providing a unified criterion for attack design and evaluation.
- Extensive experiments across multiple datasets and architectures show that our framework generally outperforms state-of-the-art VFL backdoor methods in both effectiveness and stealthiness, and remains robust against practical defenses.

II. PRELIMINARIES

A. Vertical Federated Learning

VFL aims to fully leverage features of the same users across multiple parties and exploit the correlations among them. In this setting, one party, called the active party, owns the labels Y , while the other parties, called passive parties, hold only local features. For example, given K passive parties, each party possesses a subset of user features without the corresponding labels, denoted as $D = (D_1, D_2, \dots, D_K)$.

Depending on the model architecture, VFL can be divided into two types: VFL without model splitting and VFL with model splitting [10]. In VFL without model splitting, the active party does not train a model and can only perform simple operations, such as computing the sum or average of the results from the passive parties. Each passive party produces a local prediction, while the active party merely aggregates these predictions and cannot adjust the contributions of different parties. In contrast, VFL with model splitting, as illustrated in Fig. 1, divides the model into two parts: the bottom models of the passive parties

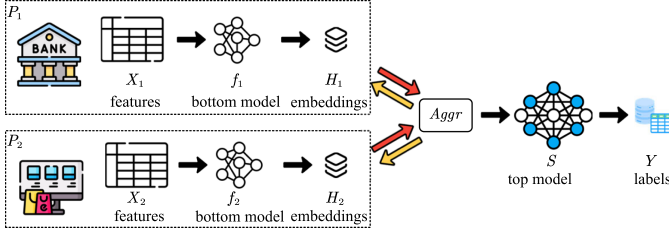


Fig. 1. Illustration of vertical federated learning with model splitting.

and the top model of the active party. The active party collects information from all passive parties and uses its top model to aggregate the results. This design enables more accurate predictions than those produced by the passive parties alone, which also makes backdoor attacks more challenging. In this paper, we focus on VFL with model splitting, as it represents a more general framework and presents greater challenges.

In VFL, each training iteration consists of three steps. First, each passive party P_i extracts local embeddings H_i using its bottom model f_i , i.e., $H_i = f_i(X_i; \theta_i)$, where θ_i denotes the model parameters and X_i represents the subset of features held by P_i . Second, the passive parties send their embeddings to the active party, which processes them using its top model S and the aggregation method $Aggr$, i.e., $\hat{Y} = S(Aggr\{H_1, H_2, \dots, H_K\})$. Third, the active party computes gradients with respect to its model parameters and the received embeddings, then sends the latter back to the passive parties to continue gradient backpropagation. These steps repeat iteratively until the entire model converges. During inference, for a given input sample x , each passive party computes its local embedding h_i and sends it to the active party. The active party then uses these embeddings to produce the prediction $\hat{y}$.

B. Threat Model

In this paper, we focus on scenarios where the attacker is a passive party, as attacks by the active party are less motivated and considerably easier to defend against. We assume there are $K + 1$ parties, where P_{K+1} is the active party and the others, $P_1, P_2, \dots, P_K$, are passive parties. Among the passive parties, we designate P_{adv} as the attacker.

a) Attacker's goal: The attacker aims to inject a backdoor into the top model. When the attacker uploads a benign embedding to the active party, the top model should behave normally. However, when the attacker uploads an embedding containing a backdoor trigger, the top model should misclassify it as a predefined target class. It is important to note that such targeted attacks are more challenging than untargeted attacks, where any incorrect class is considered a successful outcome.

b) Attacker's knowledge: The attacker has a local dataset D_{adv} containing partial features of the training samples and uses its bottom model f_{adv} to extract embeddings from this data. During training, we assume the attacker possesses a small set of labeled samples from both the target class and non-target classes. In VFL, passive parties are expected to hold valuable data to qualify for collaboration. Therefore, it is reasonable to assume that an attacker can infer the labels of a small number of samples

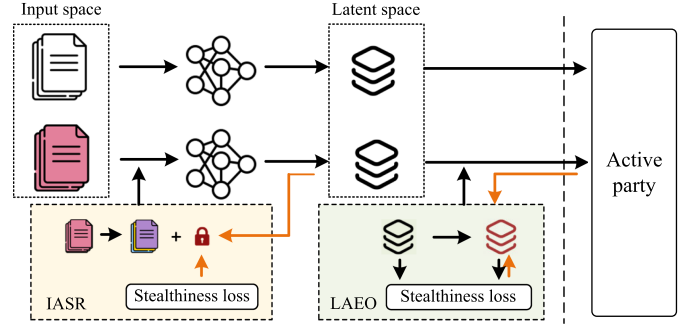


Fig. 2. Overview of our attack methods. IASR: input-space attack via sample replacement. LAEO: latent-space attack via embedding optimization.

using its local data [11]. For example, consider a collaboration between a bank and an e-commerce company for credit assessment, where the e-commerce company acts as the attacker. The company aims to classify the creditworthiness of certain users as good. Before the collaboration, it can perform independent credit assessments using its own data. Although these assessments may be less accurate than those obtained through collaboration, identifying some users with varying creditworthiness should be relatively straightforward.

c) Attacker's capability: In VFL, the attacker can manipulate its local dataset D_{adv} , its local model f_{adv} , and, if desired, the embeddings H_{adv} that it uploads. The attacker can also choose when to start and stop the attack. Crucially, the attacker cannot access or modify the labels or models of other parties.

III. METHODOLOGY

In this section, we present overview of methodology and detail each step.

A. Overview

As shown in Fig. 2, our framework leverages the protocol-exposed interfaces in split-model VFL. In this setting, a compliant adversary can influence the global model only through gradient-mediated interactions. We identify two complementary attack surfaces. The first is the input space, where the adversary manipulates local features. The second is the latent space, where the adversary controls the transmitted embeddings.

Unlike prior works that treat these spaces in isolation, we formulate backdoor injection as a unified optimization problem. The attacker optimizes a joint objective:

$$\min \mathcal{L}_{atk} + \mathcal{L}_{stealth} \quad (1)$$

Here, $\mathcal{L}_{atk}$ encourages the association between the backdoor trigger in input space or malicious embedding in latent space and the target class, while $\mathcal{L}_{stealth}$ enforces statistical consistency between poisoned and benign samples.

Strategic Complementarity of Dual-Space Attacks: The dual-space approach balances stealth at the data source with potency at the representation level. The first mechanism, Input-space Attack via Sample Replacement (IASR), works on local input features. It leverages the gradient-coupling effect to steer the shared model during backpropagation. The perturbations pass

through the bottom model's non-linear transformations, producing embeddings that appear natural. This allows IASR to bypass proactive anomaly detection in the latent space. The second mechanism, Latent-space Attack via Embedding Optimization (LAEO), directly manipulates the transmitted embeddings. It is architecture-agnostic and achieves higher attack success rates by bypassing input-space propagation constraints. Together, these two mechanisms ensure robustness across defenses. IASR provides stealth for systems with strict latent inspection, while LAEO delivers strong impact against complex VFL architectures.

Strategic Applicability of the Dual-Space Attacks: If the adversary possesses the same level of knowledge, the choice between these two strategies depends on the specific security deployment of the VFL system. IASR (input-space) is best deployed when the active party employs extremely strict anomaly detection directly on the latent payload interface. By manipulating local features, IASR forces the perturbations through the bottom model's non-linear transformations, potentially making them less conspicuous to direct latent payload inspection. Conversely, LAEO (latent-space) is superior when the adversary has direct programmatic control over the transmitted payload. LAEO is architecture-agnostic, bypassing input-space propagation constraints to achieve higher attack success rates and more robust stealth control.

B. Input-Space Attack via Sample Replacement

In VFL, the active party controls the labels, making it impossible for the attacker to modify them directly. To bypass this restriction, the attacker can manipulate its local data, which consists of a subset of all available features, denoted as $x^i_{adv} \in x^i = \{x^i_1, x^i_2, \dots, x^i_k\}$.

In this subsection, we introduce the *input-space attack via sample replacement (IASR)*. IASR combines input sample replacement with gradient-driven feature manipulation, inducing implicit correlations between local feature patterns and the target decision outcome. Throughout this process, the attacker neither accesses nor modifies ground-truth labels; all learning signals are generated by the standard VFL training protocol. During training, we also optimize the trigger to strengthen the attack. We describe the method as follows.

a) Step 1: For a given training instance associated with the target decision outcome at the active party, the attacker replaces its local feature realization with a randomly selected sample x^i_{adv} from a different class. From the attacker's perspective, this operation is performed without knowledge of the true label, and the modified sample remains fully compliant with the VFL training protocol, i.e.,

$$h^t_{adv} = f_{adv}(\tilde{x}^i_{adv}; \theta_{adv}), \quad (2)$$

and

$$\tilde{x}^i_{adv} = x^i_{adv} + M \odot T, \quad (3)$$

where M is the trigger mask, T is the input-space trigger, and $\odot$ denotes the Hadamard product. By replacing the input sample, the attacker introduces a controlled mismatch between local

Algorithm 1: Input-Space Attack via Sample Replacement.

Input: Target decision instance x^i_{target} , random non-target sample x^i_{adv} , trigger mask M , initial trigger T , learning rate η , amplification factor γ , constraint weight λ

Output: Optimized bottom model θ_{adv} and trigger T

1: **for** each training iteration **do**

2: **Step 1: Sample Replacement & Forward Propagation**

3: $\tilde{x}^i_{adv} \leftarrow x^i_{adv} + M \odot T$

4: $h^t_{adv} \leftarrow f_{adv}(\tilde{x}^i_{adv}; \theta_{adv})$

5: Send h^t_{adv} to the active party and receive gradient g^t_{adv}

6: **Step 2: Model Parameter Update**

7: $\nabla_{\theta_{adv}} L \leftarrow g^t_{adv} \cdot \frac{\partial h^t_{adv}}{\partial \theta_{adv}}$

8: Update f_{adv} using $\nabla_{\theta_{adv}} L$

9: **Step 3: Trigger Optimization with Stealth Regulation**

10: $\nabla_T L \leftarrow \gamma \cdot g^t_{adv} \cdot \frac{\partial h^t_{adv}}{\partial T} + \lambda \cdot \frac{\partial \|T\|_2}{\partial T}$

11: $T \leftarrow T - \eta \cdot \nabla_T L$

12: **end for**

feature realizations and the corresponding training signals generated by the active party. Importantly, this process does not involve modifying any ground-truth labels. Instead, the backdoor effect emerges implicitly through gradient-based optimization, which causes the top model to associate the injected trigger pattern with the target decision outcome over the course of training.

b) Step 2: During backpropagation, the attacker computes the gradient of its bottom model based on the received gradient as:

$$\nabla_{\theta_{adv}} L = g^t_{adv} \cdot \frac{\partial h^t_{adv}}{\partial \theta_{adv}}, \quad (4)$$

where g^t_{adv} is the gradient received from the active party. The attacker then uses $\nabla_{\theta_{adv}} L$ to update the parameters of its bottom model f_{adv} .

c) Step 3: In addition to model parameters, the backdoor trigger T can also be optimized by using the received gradient g^t_{adv} . However, doing so tends to increase the magnitude of T , resulting in a significant change to the readable data and making the backdoor attack unstealthy. Therefore, we further introduce a penalty term to limit the magnitude of T . Let γ and λ be adjustable coefficients. The backdoor trigger is optimized as:

$$\nabla_T L = \gamma \cdot g^t_{adv} \cdot \frac{\partial h^t_{adv}}{\partial T} + \lambda \cdot \frac{\partial \|T\|_2}{\partial T}, \quad (5)$$

where γ amplifies the received gradients to accelerate the update of the trigger and λ adjusts the level of constraint. The detailed algorithm of IASR is displayed in Algorithm 1.

C. Latent-Space Attack via Embedding Optimization

When the attacker can directly modify its latent embeddings, performing attacks in the latent space can be more effective than in the input space, as the embeddings are closer to the top model.

For this scenario, we propose *latent-space attack via embedding optimization (LAEO)*, which leverages the gradient

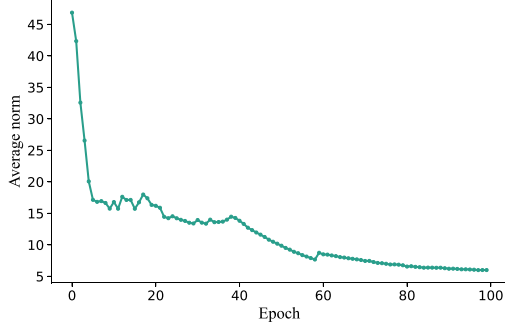


Fig. 3. Average ℓ^2 -norm of embeddings during model training. As training progresses, the average norm gradually decreases.

information returned by the active party during training to craft malicious embeddings. This approach enables not only more effective backdoor insertion but also improved stealthiness—an aspect that has received little attention in prior work. Our preliminary experiments in Fig. 3 reveal that the average norm of embeddings sent to the active party steadily decreases over the course of training, likely due to regularization techniques such as weight decay. As a consequence, without a proper optimization mechanism, backdoor attacks executed during training may generate malicious embeddings whose magnitudes deviate substantially from those of normal embeddings. Such anomalies (e.g., excessively large norms) make the attack easily detectable by the active party.

To achieve both effectiveness and stealthiness, we design LAEO so that the resulting malicious embeddings (1) mislead the top model regardless of the embeddings from other parties, and (2) maintain high stealthiness, i.e., their distribution closely resembles that of clean embeddings. The detailed algorithm of LAEO is displayed in Algorithm 2.

To satisfy the first requirement, LAEO replaces the embeddings of samples from D_{target} with malicious embedding V to establish a correlation between the V and the target label, where D_{target} is a small set of samples from the target class known to the attacker. LAEO then optimizes V during training to further improve its effectiveness and stealthiness.

Concretely, to adapt the malicious embedding to the evolving distribution of clean embeddings, LAEO employs a deviation loss,

$$L_{dev} = \min_{x_{adv}^i \in D_{target}} \|V - f_{adv}(x_{adv}^i; \theta_{adv})\|_2, \quad (6)$$

which is defined as the minimum ℓ^2 distance between V and the embedding of a target-class sample. Combined with the original loss of the active party,

$$L_{ori} = L(S(\text{Aggr}\{H_1, H_2, \dots, H_K\}), Y), \quad (7)$$

we obtain the total loss for optimizing V :

$$L_{tot} = L_{ori} + \lambda \cdot L_{dev}, \quad (8)$$

where λ is a coefficient that controls the weight of the deviation loss.

As shown in (6), the deviation loss is independent of the active party and other parties, and thus can be optimized by

Algorithm 2: Latent-Space Attack via Embedding Optimization.

Input: number of training epochs E , known target-class samples D_{target} , model learning rate α , embedding learning rate η , amplification factor γ , and weighting constant λ

Output: a malicious embedding V

1: Use a random target-class sample to initialize V .

2: **for** $epoch = 1$ to E **do**

3: **for** each batch X_{adv} **do**

4: $H_{adv} \leftarrow f_{adv}(X_{adv}; \theta_{adv})$

5: **for** $x_{adv}^i \in X_{adv}$ **do**

6: **if** $x_{adv}^i \in D_{target}$ **then**

7: $h_{adv}^i \leftarrow V$

8: **end if**

9: **end for**

10: Send embeddings H_{adv} to the active party and receive gradients g_{adv} .

11: $g_{ori} \leftarrow 0$

12: **for** $x_{adv}^i \in X_{adv}$ **do**

13: **if** $x_{adv}^i \in D_{target}$ **then**

14: $g_{ori} \leftarrow g_{ori} + g_{adv}^i$

15: **end if**

16: **end for**

17: $L_{dev} \leftarrow \min_{x_{adv}^i \in D_{target}} \|V - f_{adv}(x_{adv}^i; \theta_{adv})\|_2$

18: $g_{dev} \leftarrow \nabla_V L_{dev}$

19: $g \leftarrow \gamma \cdot g_{ori} + \lambda \cdot g_{dev}$

20: $V \leftarrow V - \eta \cdot g$

21: Update model f_{adv} with g_{adv} and α .

22: **end for**

23: **end for**

the attacker alone. As shown in (9), the derivation of the overall loss with respect to the malicious embedding V comprises two parts: the gradients returned by the active party, and the gradients calculated by the attacker locally. To enhance the attack performance, we amplify the received gradients by a factor of γ . Subsequently, derive the malicious embedding, as detailed in lines 11-18 of Algorithm 2. The final update of the malicious embedding depends on two components, the attack objective and the constraint objective, as detailed in line 19 of Algorithm 2. Finally, the learning rate η is used to update V .

$$\nabla_V L_{tot} = \gamma \cdot \frac{\partial L_{ori}}{\partial V} + \lambda \cdot \frac{\partial L_{dev}}{\partial V} = \gamma \cdot g_{ori} + \lambda \cdot g_{dev}. \quad (9)$$

Existing approaches often rely on a surrogate model to approximate the classification boundary of the top model and optimize triggers by using the gradients of the surrogate model [3], [4]. However, building surrogate models generally requires samples from all classes and additional training. Notably, these gradients can also be constructed during training. Compared to these methods, our approach eliminates the need for surrogate models, simplifies the process, and enhances the attack's efficiency by directly optimizing the embedding in the latent space during training.

TABLE II
ATTACK PERFORMANCE OF OUR METHODS COMPARED WITH BASELINES. BA (↑): BENIGN ACCURACY. ASR (↑): ATTACK SUCCESS RATE. AS (↓): ATTACK STEALTHINESS

Dataset	Metric	Clean	LR-BA	VILLAIN	SDDP	IASR	LAEO
CIFAR-10	BA (%)	86.88	83.81	85.24	85.61	85.74	85.19
	ASR (%)	1.40	11.25	87.35	88.65	94.50	96.00
	AS	-	0.696	1.360	0.547	0.846	0.022
CIFAR-100	BA (%)	84.67	82.4	81.37	82.45	81.98	82.38
	ASR (%)	4.60	14.7	30.65	95.35	100.00	98.45
	AS	-	0.915	1.161	0.527	0.882	0.146
CINIC-10	BA (%)	80.52	77.98	78.66	78.83	79.26	79.13
	ASR (%)	1.55	13.35	90.75	97.50	95.20	99.25
	AS	-	0.707	0.742	0.426	0.976	0.057
BHI	BA (%)	89.45	86.12	87.35	87.90	88.15	87.84
	ASR (%)	1.05	12.40	89.50	92.15	96.40	97.85
	AS	-	0.625	0.985	0.482	0.790	0.038
BM	BA (%)	83.15	80.24	81.50	81.95	81.42	81.70
	ASR (%)	0.80	9.65	72.40	86.10	91.55	94.30
	AS	-	0.812	1.240	0.510	0.915	0.086

D. Stealthiness Considerations in VFL

In vertical federated learning, each party keeps its raw data private and transmits only latent embeddings to the active party. Consequently, a successful backdoor attack must ensure that its malicious embeddings remain indistinguishable from benign ones in the embedding space; otherwise, they can be easily flagged by standard server-side defenses. Stealthiness therefore becomes a fundamental design requirement for backdoor attacks in VFL.

To quantify stealthiness in a principled yet practical manner, we introduce an *attack stealthiness* (AS) metric that captures two minimal consistency requirements in the VFL setting. We do not claim AS to be an exhaustive or theoretically optimal measure of stealthiness. Instead, it is designed to reflect whether poisoned embeddings remain sufficiently consistent with benign embedding dynamics to evade common server-side statistical inspections. Specifically, AS captures two aspects: (i) source consistency, which measures the similarity between poisoned embeddings and the embeddings of clean samples from the same class, and (ii) target consistency, which measures the similarity between poisoned embeddings and the embeddings of target-class samples known to the attacker. Formally, given a poisoned dataset D_p :

$$AS = \sum_{x_{adv}^i \in D_p} \left[\frac{1}{2} \min_{s \in E_{adv}^y} \frac{\|h_{adv}^i - s\|_2}{E[\|s\|_2]} + \frac{1}{2} \min_{z \in Z_{adv}^{\hat{y}}} \frac{\|h_{adv}^i - z\|_2}{E[\|z\|_2]} \right] \quad (10)$$

where h_{adv}^i is the embedding of sample x_{adv}^i from the attacker, y is its true label, E_{adv}^y contains clean embeddings of class y , $\hat{y}$ is the predicted label, and $Z_{adv}^{\hat{y}}$ contains embeddings predicted as $\hat{y}$. A lower AS indicates that the attacker's embeddings are more consistent with the model's existing embedding distribution and knowledge, reducing detectability.

We use minimum-distance statistics in (10) because they capture the local proximity between poisoned and benign embeddings. This proximity is important for evading distance-based

outlier detectors. In contrast, mean distances are sensitive to the multimodal structure of multi-class embeddings and can hide local anomalies. However, AS mainly measures ℓ^2 -based statistical proximity and may miss subtle semantic anomalies. This metric directly guides the optimization of our latent-space attack. By adding a stealthiness regularization term derived from AS, the attack encourages the optimized embeddings to achieve the target label association while maintaining high stealthiness. This design helps the malicious embeddings evade standard defenses and maximizes the overall attack success.

IV. EXPERIMENTS

A. Experimental Setup

Datasets: We evaluate our methods on three image classification datasets: CIFAR-10, CIFAR-100 [12], and CINIC-10 [13]. To further assess generalizability beyond image-based tasks, we additionally evaluate on two datasets representative of real-world VFL deployment scenarios: the Breast Histopathology Images (BHI) dataset [14] (healthcare) and the Bank Marketing (BM) dataset [15] (finance). For tabular data, each party holds a disjoint subset of features, following the standard VFL vertical partitioning protocol. This selection allows us to assess the effectiveness and robustness of our approach across diverse scenarios, including datasets with different numbers of classes (e.g., CIFAR-10 versus CIFAR-100) and varying amounts of training data (e.g., CIFAR-10 versus CINIC-10). To ensure balanced data allocation among passive parties, we evenly distribute input features for each dataset. For instance, in a two-party setting, each image is split vertically in half, with each party receiving one half [3], [11], [16], [17].

Model architectures: We adopt a partial VGG-16 architecture for the bottom models and enhance the top model by replacing the fully connected layers with a convolutional layer, two ResNet blocks, and a linear layer. This design improves model accuracy compared with traditional configurations (see Table II). In existing studies [3], [5], [16], researchers typically use VGG-16 [18] or ResNet-20 [19] for the bottom models, while the top model usually consists of one or more fully connected layers. However, this configuration often yields lower accuracies, e.g., below 85% on CIFAR-10 and 75% on CINIC-10. Our design addresses this limitation while remaining compatible with the VFL framework.

Attack settings: We conduct experiments in a basic attack scenario with two passive parties, each holding half of the input features. The top model uses embedding concatenation as the default aggregation method. We set the poisoning rate to 1% for CIFAR-10 and CINIC-10, and 0.1% for CIFAR-100. The attacker has the same number of labeled target-class samples as poisoned samples. Default hyperparameters are set as $\gamma = 2$, $\lambda = 1$, and $\eta = 0.01$. For IASR, additional samples from non-target classes are included, with their number equal to the labeled target-class samples. We adopt this configuration to create a balanced and controlled experimental environment. The two-party setting with evenly split features represents a simple yet standard scenario for VFL experiments. The chosen poisoning rates and labeled samples ensure that attacks are measurable while reflecting practical conditions. Default hyperparameter values

are selected to balance attack effectiveness and stealthiness, and the inclusion of additional non-target samples for IASR supports proper execution of the input-space attack.

Metrics: We evaluate model performance using benign accuracy (BA) and attack success rate (ASR). BA denotes the accuracy of the infected model on clean test samples, while ASR measures the proportion of successful attacks, i.e., test samples misclassified into the target class. For CIFAR-100, where the number of classes is larger, we report top-5 BA and ASR. In addition, we adopt the attack stealthiness (AS) metric defined in (10) to quantify the stealthiness of backdoor attacks.

Baseline methods: We choose three backdoor attack methods as baselines: LR-BA [3], a backdoor attack at inference time; VILLAIN [16], a latent-space backdoor attack with static triggers; and SDDP [11], a latent-space attack with dynamic triggers. For fair comparison, we use the same evaluation protocol for all methods, including data preprocessing, attacker capabilities, and model architectures. For LR-BA, we use the same number of labeled samples for CIFAR-10 and CINIC-10 as in our methods, while maintaining its original setting for CIFAR-100, which requires more labeled samples than ours. For VILLAIN, we omit the label inference process and apply only its backdoor attack method.

We note that the performance of VILLAIN in our evaluation (e.g., 87.35% ASR on CIFAR-10) is lower than the 98% ASR reported in its original paper. This discrepancy stems primarily from the top model architecture. While the original VILLAIN paper employs simple fully connected layers for the top model, our evaluation utilizes a more complex and realistic architecture (convolutional layers and ResNet blocks). Such non-linear complexity natively resists static triggers more effectively, leading to the observed ASR drop. Furthermore, although the label inference component of VILLAIN was omitted, we purposefully aligned the auxiliary knowledge assumptions (e.g., the number of available labeled samples) across all baseline methods. This alignment ensures a strictly fair, apples-to-apples comparison focused solely on the efficacy of the underlying backdoor injection mechanisms.

Experimental environment: All experiments are performed on a workstation with Intel(R) Xeon(R) Gold 6133 CPU @ 2.50 GHz and NVIDIA GeForce RTX 4090 GPUs. The proposed methods are implemented with PyTorch.

B. Attack Performance

Overall performance: Table II reports the results of our methods compared with baselines on two-party VFL. Our methods outperform the baselines, achieving higher attack effectiveness and improved stealthiness. Dynamic attack methods, including SDDP, IASR, and LAEO, consistently outperform the static attack method VILLAIN, especially on CIFAR-100. The performance of VILLAIN largely depends on the magnitude of its trigger relative to that of clean embeddings. As shown in Fig. 3, the static trigger requires a prolonged training period to reach a higher magnitude. This requirement is impractical, as the active party is unlikely to extend training time without corresponding improvements in model accuracy. Comparing input-space and

TABLE III
EVALUATION IN MULTI-PARTICIPANT SCENARIOS. K : THE NUMBER OF PASSIVE PARTIES.

Dataset	Attack	$K=4$			$K=8$		
		BA (%)	ASR (%)	AS	BA (%)	ASR (%)	AS
CIFAR-10	Clean	81.53	2.05	-	78.08	3.35	-
	VILLAIN	79.21	74.90	1.061	75.39	71.80	0.828
	SDDP	80.34	59.65	0.675	76.77	40.95	0.669
	IASR	80.75	86.50	0.872	76.33	78.55	1.074
	LAEO	80.03	94.10	0.030	77.49	92.55	0.159
CIFAR-100	Clean	80.60	5.50	-	77.64	4.40	-
	VILLAIN	76.57	17.60	1.035	74.08	31.85	1.054
	SDDP	77.77	83.55	0.884	74.99	73.30	0.822
	IASR	77.78	99.10	1.039	75.38	99.80	1.114
	LAEO	77.77	100.00	0.201	75.58	100.00	0.239
CINIC-10	Clean	75.78	2.35	-	73.49	1.90	-
	VILLAIN	74.65	95.55	0.713	71.16	100.00	0.732
	SDDP	74.78	92.65	0.602	71.45	97.90	0.759
	IASR	74.22	95.70	0.931	71.09	79.45	1.111
	LAEO	74.23	95.75	0.136	71.50	97.90	0.246

latent-space attacks, both approaches achieve satisfactory effectiveness. However, latent-space attacks, particularly SDDP and LAEO, exhibit superior stealthiness because they can directly constrain the embeddings uploaded to the server. Among latent-space attacks, LAEO outperforms SDDP in both effectiveness and stealthiness, thanks to its embedding optimization mechanism combined with a stealthiness constraint.

In addition, IASR outperforms LR-BA [3], an input-space attack that generates optimized embeddings using a local surrogate model. LR-BA achieves suboptimal ASR because it struggles to emulate complex top models; its surrogate attains less than 50% accuracy on CIFAR-10. Moreover, LR-BA is computationally inefficient. It requires a separate, resource-intensive pre-training phase to build the surrogate model before launching the attack. In contrast, our framework operates online. It optimizes triggers and embeddings directly through the intrinsic gradient flow of VFL training. Although our dual-space optimization adds a small overhead for computing the stealthiness loss, it eliminates the costly need to train and maintain a high-fidelity surrogate model. Considering both attack effectiveness and computational efficiency, we exclude LR-BA from detailed comparisons and focus on VILLAIN and SDDP.

The generalizability of our framework is further confirmed on tabular datasets. As shown in Table II, both IASR and LAEO achieve consistently high ASR on the BHI and BM datasets while maintaining BA close to the clean baseline. Notably, LAEO retains its stealthiness advantage with substantially lower AS values compared to VILLAIN and SDDP, demonstrating that the dual-space attack framework is data-modality agnostic.

Multi-party scenario: We evaluate the attacks in scenarios with 4 and 8 passive parties. As shown in Table III, BA decreases consistently as the number of parties K increases. This trend occurs because a larger K makes it more difficult for each bottom model to learn effectively.

Nevertheless, IASR and LAEO are less affected by the number of passive parties, while the effectiveness of SDDP on CIFAR-10 and CIFAR-100 decreases noticeably as K increases. In the scenario with 8 passive parties, LAEO outperforms the other methods across nearly all metrics, demonstrating its effectiveness in multi-party settings. Interestingly, on CIFAR-10

TABLE IV
EVALUATION ON ELEMENT-WISE SUM AGGREGATION

Dataset	Metric	Clean	VILLAIN	SDDP	IASR	LAEO
CIFAR-10	BA (%)	85.39	83.57	83.84	83.96	83.67
	ASR (%)	0.50	54.75	91.75	96.10	92.15
	AS	-	0.963	0.490	0.878	0.138
CIFAR-100	BA (%)	81.85	79.36	80.67	80.24	81.19
	ASR (%)	2.10	26.20	95.25	100.00	99.90
	AS	-	1.230	0.684	1.099	0.157
CINIC-10	BA (%)	78.66	77.05	77.00	77.06	77.18
	ASR (%)	1.60	81.50	98.85	88.15	98.75
	AS	-	0.826	0.480	1.263	0.130

TABLE V
EVALUATION ON ELEMENT-WISE AVERAGE AGGREGATION

Dataset	Metric	Clean	VILLAIN	SDDP	IASR	LAEO
CIFAR-10	BA (%)	86.23	83.68	84.08	83.92	83.94
	ASR (%)	0.40	42.20	89.2	90.75	96.05
	AS	-	1.059	0.510	0.839	0.114
CIFAR-100	BA (%)	83.08	80.02	80.39	80.47	80.32
	ASR (%)	5.10	27.95	97.00	99.75	99.70
	AS	-	1.234	0.664	0.938	0.141
CINIC-10	BA (%)	78.37	77.04	77.35	77.13	77.44
	ASR (%)	2.10	76.65	98.70	78.60	93.95
	AS	-	0.804	0.501	1.242	0.047

and CINIC-10 with 8 parties, LAEO significantly surpasses IASR, highlighting the advantage of latent-space attacks over input-space attacks. In addition, we observe that VILLAIN generally achieves lower BA compared to the other methods.

Different embedding aggregation methods: While we use concatenation as the default aggregation method, the active party may employ various mechanisms to combine the received data [20]. To evaluate the impact of different aggregation methods on attack performance, we test the methods with element-wise sum aggregation and present the results in Table IV. Results for element-wise average aggregation are shown in Table V. The results indicate that LAEO performs consistently across aggregation methods, achieving the best or highly competitive performance. IASR outperforms the baselines on CIFAR-10 and CIFAR-100 but shows lower stealthiness due to the inherent nature of input-space attacks. The effectiveness of VILLAIN decreases substantially under both aggregation methods and does not excel in any of the three metrics. SDDP performs better on CINIC-10; however, its BA and AS remain inferior to those of LAEO. Overall, our methods, particularly LAEO, maintain strong effectiveness regardless of the aggregation method.

Impact of poisoning rate: Poisoning rate refers to the proportion of backdoor samples in the full dataset. A low poisoning rate combined with a high ASR typically indicates an effective attack. As shown in Fig. 4, with a small poisoning rate of 0.5% or 0.05%, IASR achieves an ASR exceeding 84%, while LAEO exceeds 95%. Moreover, our methods maintain consistent performance across different poisoning rates. Although the performance of IASR on CINIC-10 is somewhat affected by the poisoning rate, its BA remains higher than that of the other

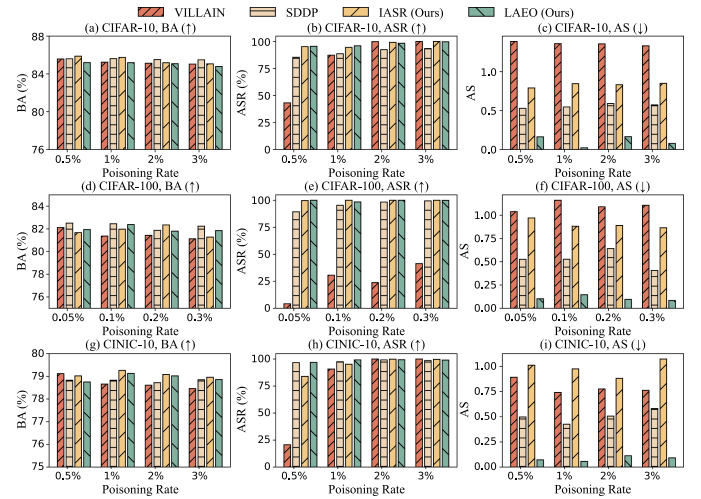


Fig. 4. Impact of poisoning rate on benign accuracy (BA), attack success rate (ASR) and attack stealthiness (AS).

TABLE VI
IMPACT OF λ ON IASR FOR BALANCING ASR ($\uparrow$) AND AS ($\downarrow$) IN CIFAR-10

Metric	λ					
	0	0.002	0.004	0.006	0.008	0.01
BA (%)	85.56	85.08	85.87	85.33	85.40	85.74
ASR (%)	100.00	100.00	98.45	97.25	95.70	94.50
AS	1.197	0.833	0.874	0.821	0.853	0.846

TABLE VII
IMPACT OF λ ON LAEO FOR BALANCING ASR ($\uparrow$) AND AS ($\downarrow$) IN CIFAR-10

Metric	λ					
	0	0.2	0.4	0.6	0.8	1.0
BA (%)	85.57	85.12	85.35	85.67	85.49	85.19
ASR (%)	100.00	99.20	97.55	96.65	96.05	96.00
AS	2.162	0.093	0.080	0.082	0.135	0.022

methods. In contrast, VILLAIN shows significant performance fluctuations under varying poisoning rates and exhibits less consistency.

Tradeoff between effectiveness and stealthiness: For IASR and LAEO, the hyperparameter λ plays a key role in balancing attack effectiveness and stealthiness. A larger value of λ emphasizes stealthiness over effectiveness, as reflected in the results shown in Tables VI and VII. In contrast, setting λ to 0 represents an extreme case, which achieves an ASR of 100% but results in a high AS, indicating lower stealthiness.

Ablation study of optimization: We compare the performance of our attack methods with and without optimization. As shown in Fig. 5, optimization substantially improves both the effectiveness and stealthiness of LAEO, increasing ASR by up to 80%. For IASR, optimization results in a slightly lower ASR, but it significantly enhances stealthiness as measured by AS. Overall, these results highlight the critical role of the optimization mechanism.

Impact of perturbation: While the embeddings used by LAEO are optimized during training, they are fixed during inference, which can be exploited by the active party for detection. To mitigate this vulnerability, we introduce perturbations into the

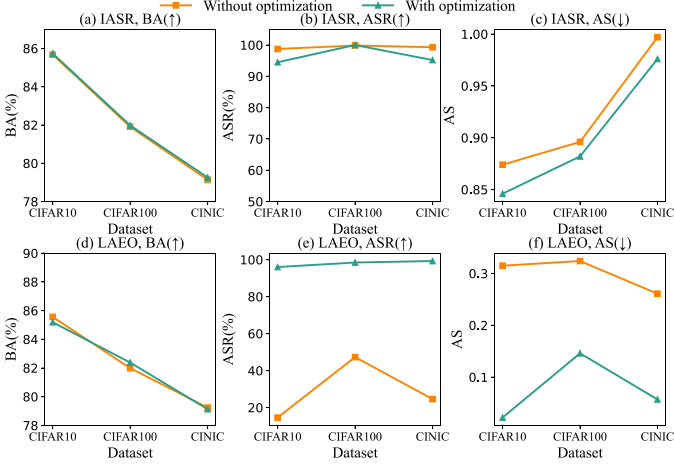


Fig. 5. Ablation study of optimization mechanism across different datasets.

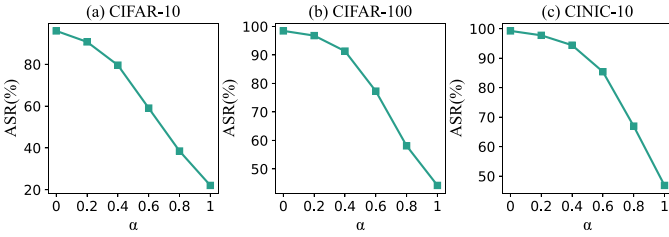


Fig. 6. Impact of perturbations on ASR of LAEO.

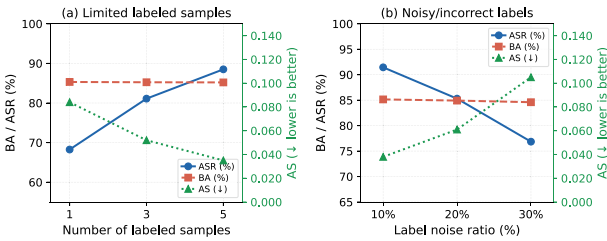


Fig. 7. Robustness analysis of LAEO on CIFAR-10. (a) Performance under extremely limited target-class supervision. (b) Performance under noisy label conditions.

malicious embeddings at inference time. Specifically, denoting the malicious embedding by V , we scale the ℓ^2 norm of the original embedding, $f_{adv}(x_{adv}^i; \theta_{adv})$, to $\alpha \|V\|_2$ and apply it as an additive perturbation, where α is a hyperparameter to control the magnitude of the perturbation. As demonstrated in Fig. 6, while an increased perturbation level leads to a decrease in ASR, LAEO remains effective for α below 0.6. At $\alpha = 1$, the attack becomes ineffective, with an ASR below 25% on CIFAR-10, due to the interference of the perturbation.

Performance under extremely limited labeled samples: To investigate the framework’s behavior under severe supervision constraints, we restricted the adversary’s available target-class samples to 1, 3, and 5 instances respectively. As shown in Fig. 7(a), while ASR decreases relative to the full-supervision baseline, the attack remains highly effective across all settings. Notably, even with a single labeled sample, LAEO achieves an ASR of 68.25%. This rapidly improves to 88.50% with only 5 samples, demonstrating that the gradient-coupling mechanism

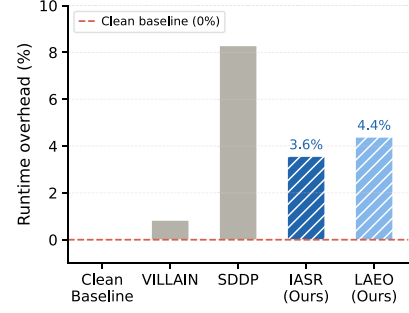
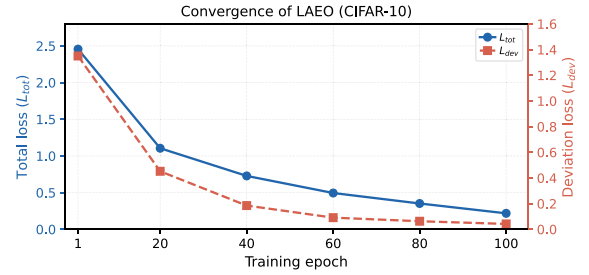


Fig. 8. Runtime overhead per training epoch on CIFAR-10 relative to the clean baseline.

Fig. 9. Empirical convergence of L_{tot} and L_{dev} for LAEO over 100 training epochs.

in VFL efficiently amplifies even minimal supervisory signals to sustain effective backdoor injection. Meanwhile, BA remains stable throughout, and AS decreases monotonically with sample count, confirming tighter embedding alignment under richer supervision.

Robustness against noisy labels: We further evaluated LAEO’s robustness when a fraction of the adversary’s inferred target labels are incorrect. As shown in Fig. 7(b), LAEO maintains strong attack effectiveness across all noise levels. Even when 20% of labels are incorrect, ASR remains above 85% with negligible impact on BA. We attribute this stability to the deviation loss, which anchors the malicious embedding to the target-class manifold independently of label accuracy, thereby preventing catastrophic misalignment. As expected, AS increases with noise ratio, reflecting a modest reduction in embedding conformity as label quality degrades.

Runtime Overhead and Communication Cost: Because our framework operates online directly through the intrinsic gradient flow of standard VFL training, it introduces strictly zero additional communication rounds and zero payload overhead between parties. Regarding computational overhead, the extra cost is strictly localized to the adversary’s calculation of the stealthiness deviation loss. As shown in Fig. 8, both IASR and LAEO introduce less than 5% runtime overhead relative to the clean baseline, making them practically imperceptible.

Stability and Convergence: While our dual-space framework relies on empirical gradient-based optimization, we observed highly stable convergence dynamics. As shown in Fig. 9, both the total classification loss (L_{tot}) and the deviation loss (L_{dev}) decrease monotonically over 100 training epochs without divergent oscillations. The boundary constraints imposed by the

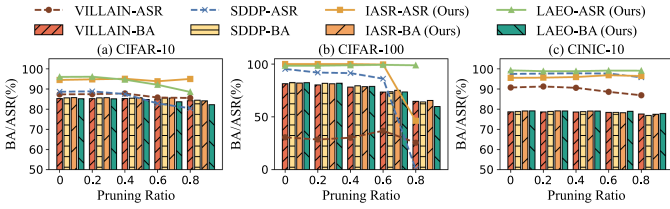


Fig. 10. Evaluation against model pruning.

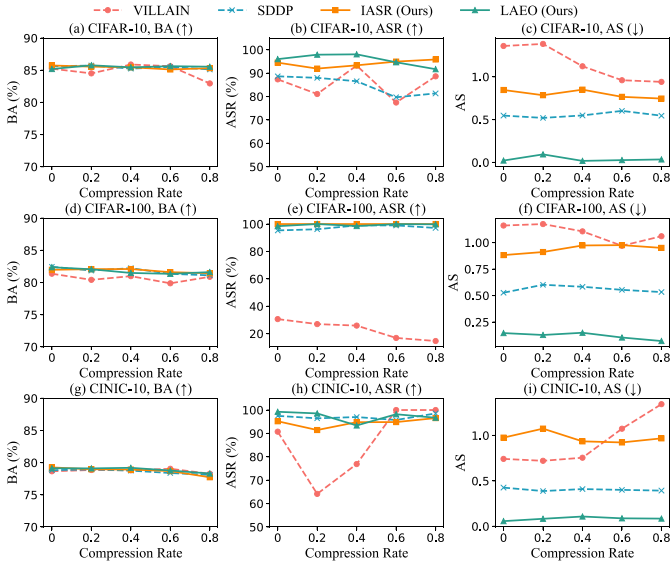


Fig. 11. Evaluation against gradient compression.

deviation loss effectively restrict malicious embeddings within the benign manifold, preventing optimization instability.

C. Robustness to Defenses

To rigorously evaluate the resilience of our proposed framework, we subject it to a diverse battery of practical VFL backdoor defenses. These encompass both training-phase mitigation techniques and inference-phase detection mechanisms. Specifically, we evaluate against Gradient Compression [21], Anti-backdoor Learning (ABL) [22], and Model Pruning [23]. Furthermore, to explicitly address representation-level stealthiness and verify robustness against statistical anomaly detection in the latent space, we evaluate against TeCo [24] and Outlier Detection. Our results demonstrate that our attacks remain highly effective across these defensive paradigms.

Gradient compression: To reduce the communication overhead of passing gradients, the active party can compress the gradients by sending only those with the largest absolute values [21]. This gradient compression may affect the optimization process of backdoor attacks. We evaluate different attacks under various compression rates, as shown in Fig. 11. The results indicate that IASR and LAEO remain largely stable, with ASR consistently above 91%. In contrast, VILLAIN shows substantial performance fluctuations in some cases, leading to lower BA and ASR overall.

Anti-backdoor learning: Anti-backdoor Learning (ABL) employs a gradient-ascent mechanism to isolate backdoor samples

TABLE VIII
EVALUATION AGAINST ABL

Dataset	Attack	DA (%)	BA (%)	ASR (%)	AS
CIFAR-10	VILLAIN	0.20	70.15	3.35	0.405
	SDDP	0.20	76.87	99.45	0.732
	IASR	0.80	77.31	82.70	0.717
	LAEO	0.00	68.88	96.50	0.011
CIFAR-100	VILLAIN	0.00	73.88	5.80	0.506
	SDDP	0.00	75.49	80.00	0.710
	IASR	0.00	65.42	98.25	0.899
	LAEO	0.00	71.51	83.65	0.162
CINIC-10	VILLAIN	4.39	59.93	7.95	0.378
	SDDP	1.17	55.39	99.95	0.378
	IASR	0.39	77.42	89.00	0.933
	LAEO	6.83	70.80	96.75	0.094

TABLE IX
EVALUATION AGAINST TeCo

Dataset	Attack	BA (%)	ASR (%)	DA(%)
CIFAR-10	VILLAIN	84.26	87.35	0.05
	SDDP	84.27	88.85	2.5
	IASR	84.77	94.55	0.65
	LAEO	84.23	95.75	0.65
CIFAR-100	VILLAIN	81.18	30.55	0.55
	SDDP	82.33	76.35	19.00
	IASR	81.95	100.00	0.00
	LAEO	81.42	97.20	1.25
CINIC-10	VILLAIN	77.90	90.65	0.85
	SDDP	78.07	97.50	0.00
	IASR	78.30	95.55	0.10
	LAEO	78.37	98.80	0.00

and disrupt the correlation between poisoned inputs and the target class [22]. As shown in Table VIII, although ABL achieves relatively low detection accuracy (DA) and substantially degrades the model's clean accuracy, it is effective in reducing the ASR of VILLAIN to below 10%. Nevertheless, IASR and LAEO remain robust against ABL, as reflected by their ASR values. Moreover, because SDDP launches its attack only after ABL has attempted to isolate backdoor samples, its ASR is similarly unaffected by the defense.

Model pruning: Model pruning [23] iteratively removes neurons in increasing order of their average activation to disable backdoors. As shown in Fig. 10, IASR and LAEO maintain more stable performance than the baseline methods across different pruning ratios, as indicated by ASR. Moreover, the decrease in ASR occurs alongside a reduction in BA, showing that model pruning cannot effectively defend against our methods without substantially degrading model performance.

TeCo: TeCo [24] is a test-time backdoor detection method based on the observation that backdoored samples show different prediction consistency under input corruptions compared to clean samples. In VFL, the active party only has access to latent embeddings, which are more sensitive to corruptions than raw input samples. As a result, the model's performance on backdoor samples and clean samples becomes similar when embeddings are corrupted, rendering TeCo ineffective in detecting backdoors, as shown in Table IX. Consequently, the attacks maintain stable performance in most cases. In particular, our methods remain highly effective against TeCo, achieving ASR values above 94.5%.

TABLE X
THE PERFORMANCE OF BACKDOOR ATTACKS UNDER OUTLIER DETECTION

Dataset	Metric	VILLAIN	SDDP	IASR	LAEO
CIFAR-10	AS	1.360	0.547	0.846	0.022
	F1 score	0.98	0.86	0.92	0.53
CIFAR-100	AS	1.161	0.527	0.882	0.146
	F1 score	0.98	0.56	0.98	0.47
CINIC-10	AS	0.742	0.426	0.988	0.057
	F1 score	0.17	0.85	0.09	0.00

Outlier detection: Standard defenses often rely on the assumption that backdoor samples form a distinct cluster or exhibit anomalous magnitudes in the latent space. We evaluate this using Median Absolute Deviation (MAD) with the ℓ^2 -norm, a robust statistical outlier detection method. As presented in Table X, our LAEO method achieves substantially lower F1 scores compared to baselines, indicating that our poisoned embeddings are statistically indistinguishable from benign ones. Crucially, this resistance to statistical outlier detection implies robustness against more advanced defenses, which fundamentally rely on separating latent representations into distinct benign and malicious clusters. Since our explicit stealth regulation minimizes the deviation loss, it effectively merges the poison distribution into the benign manifold, thereby neutralizing the segregation mechanism required by such cluster-based defenses.

D. Analysis of Attack Stealthiness

Our experiments show that the proposed AS metric reliably captures the relationship between embedding-level stealthiness and backdoor detectability in VFL. Across datasets, aggregation strategies, and attack methods, lower AS values correspond to poisoned embeddings that remain close to the benign manifold, exhibit fewer outliers, and achieve lower detection rates, while higher AS values indicate larger embedding deviations and higher detectability. The results in Tables II, IV, V, and X support this conclusion.

i) *AS reveals stealthiness beyond conventional metrics:* Benign accuracy and attack success rate (ASR) evaluate model performance and attack effectiveness, but they do not measure how poisoned embeddings deviate from the benign distribution. As shown in Table II, input-space attacks achieve competitive ASR but yield significantly higher AS values than our latent-space optimized attack. This result indicates that input-space perturbations propagate into the latent space and produce distinguishable outliers, reducing representation-level stealthiness. In contrast, our AS-aware method explicitly constrains embedding conformity and achieves much lower AS.

ii) *AS correlates with detectability under both analytical and practical defenses:* Results in Table X show that, in most cases, higher AS leads to more separable embeddings and higher F1 scores in embedding-based outlier detection. Attacks with lower AS, such as LAEO, keep embeddings closer to the benign distribution and result in lower detection rates. A similar trend appears under TeCo [24] in Table IX, where low-AS attacks preserve high ASR while evading detection.

iii) *AS remains stable across datasets and aggregation strategies:* Tables IV and V show that the AS-based ranking of attack methods remains consistent across datasets and aggregation types, even when benign accuracy and ASR vary. This stability indicates that AS captures an intrinsic property of embedding conformity rather than task-specific artifacts.

iv) *AS reflects the trade-off between attack effectiveness and stealthiness:* IASR and LAEO both achieve high ASR, but LAEO consistently produces lower AS across all settings. Input-space attacks lack direct control over latent representations and tend to generate detectable outliers with high AS. In contrast, LAEO constrains poisoned embeddings within the benign distribution and balances effectiveness with stealthiness, which AS captures effectively.

V. RELATED WORK

Backdoor attacks are commonly divided into poison-label and clean-label attacks based on label consistency [25]. Poison-label attacks inject samples with incorrect labels to create malicious associations [26], [27], [28]. Clean-label attacks keep labels unchanged but craft samples to induce misclassification [29], [30]. In practice, poison-label attacks often require lower poisoning rates.

In vertical federated learning (VFL), backdoor attacks mainly target either the input space or the latent space. Input-space attacks inject predefined or optimized triggers into local features. Existing methods include gradient-replacement attacks [31], universal clean-label attacks based on projected gradient ascent [8], target-efficient clean backdoor attacks [32], adversarial embedding fine-tuning [3], surrogate-based optimization of input-space triggers [5], and runtime label inference to strengthen trigger-label associations [33].

Latent-space attacks manipulate intermediate representations exchanged between parties. Representative approaches include replay and generation attacks on robust embeddings [4], additive triggers on high-variance embedding dimensions [16], and SplitNN-specific embedding replacement near dense latent regions [11].

Regarding stealthiness, many attacks rely on predefined triggers or coarse embedding constraints [4], [31], [33]. Some methods use large-norm embeddings to dominate benign features [3]. Others explicitly bound perturbations using ℓ^2 or ℓ^∞ norms [5], [8], [32]. More recent studies limit trigger magnitude or selected dimensions in the latent space [11], [16]. However, these methods often ignore how embedding distributions evolve during training. In contrast, our work dynamically generates class-indicative triggers in both input and latent spaces and systematically compares their effects.

VI. CONCLUSION

In this paper, we examined the security risks of vertical federated learning (VFL), with a particular focus on the largely underexplored domain of *clean-label backdoor attacks*. We proposed a comprehensive attack framework that operates in both the input space and the latent space of passive parties. By leveraging gradient information available during standard VFL training, our methods effectively implant backdoors without

modifying labels or relying on surrogate models. In addition, we introduced a principled stealthiness metric and a corresponding regularization strategy tailored to VFL's split-model architecture, substantially enhancing the concealment of backdoors against common server-side defenses. Extensive experiments across multiple datasets and VFL configurations demonstrate that the proposed framework achieves strong attack effectiveness while maintaining high stealthiness and robustness, outperforming existing methods that depend on label inference or impose no explicit constraints on triggers.

Despite these advances, our methods have several limitations. In particular, they assume that the adversary has access to a small set of auxiliary clean samples from target classes, which may not always be available in practice. Furthermore, variations in the number of participating parties and the heterogeneity of model architectures can influence attack performance. Future work may explore more adaptive and data-efficient attack strategies to mitigate these limitations, as well as further enhance the generalizability and stealth of clean-label backdoor attacks in VFL.

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